

ME5412 Robotics for Healthcare

Stroke and Stroke Rehabilitation

中风与中风康复

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OUTLINE

- Definition & Epidemiology of stroke
 - Consequences of stroke
 - Stroke recovery Mechanism
 - Clinical practices of stroke rehabilitation and assessment
 - Stroke Rehabilitation Interventions
 - Robotics in stroke rehabilitation
- 中风的定义与流行病学
 - 中风的后果
 - 中风恢复机制
 - 中风康复的临床实践与评估
 - 中风康复干预手段
 - 中风康复中的机器人技术



中风的定义与流行病学

DEFINITION & EPIDEMIOLOGY OF STROKE

中风

STROKE (ANIMATION)

大脑侧面血管解剖图。

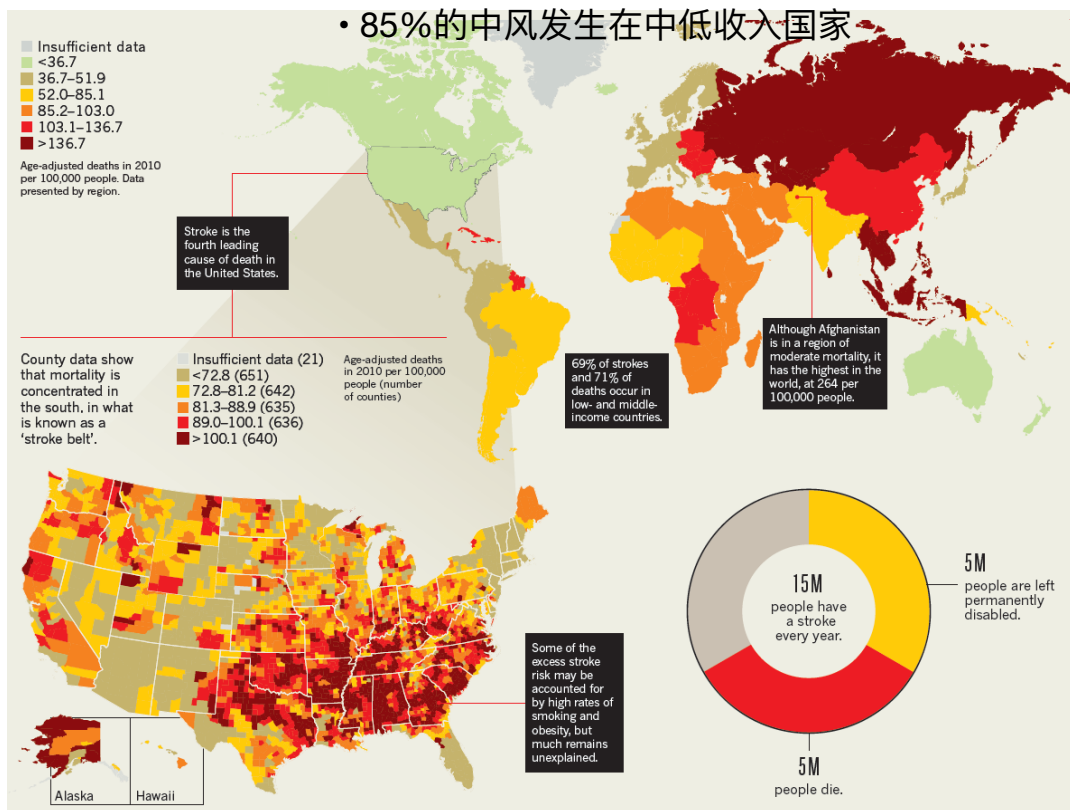


中风数据统计

STROKE STATISTICS

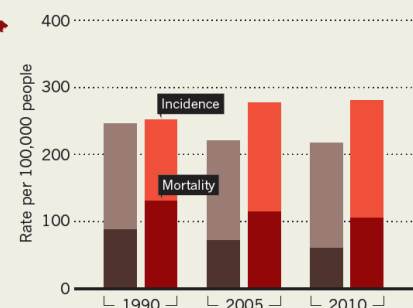
- 全球中风分布图（年龄调整死亡率/10万人）
- 85%的中风发生在中低收入国家

• 美国"中风带"（东南部死亡率最高）

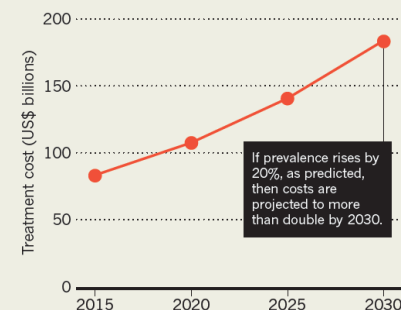


- 高收入国家发病率和死亡率下降，但患病率上升

AGE CONCERN
Both incidence and mortality are falling in high-income countries. But prevalence is rising, particularly in high-income nations.



FINANCIAL TIMES
Direct medical costs are expected to rise substantially in the United States because prevalence will increase as the population ages.



• 美国直接医疗成本预计2030年翻倍

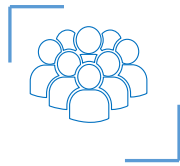
- 每年1500万人中风：500万死亡，500万永久残疾

中风是全球第二大死因，而且幸存者中很多人永久残疾——这就是康复机器人存在的理由。

关键数字

STROKE STATISTICS

- 7000万：2030年全球中风幸存者
- 1600万：全球每年新发病例
- 30–66%：中风6个月后仍有运动障碍的患者比例
- 737亿美元：2010年美国中风直接+间接经济成本



70 million

Worldwide stroke survivors by 2030¹



16 million

New cases worldwide per year²



30–66%

Patients exhibit motor deficits six months after stroke²



73.7 billion

Estimated direct and indirect cost of stroke in 2010 in the US³

1. Feigin VL, Forouzanfar MH, Krishnamurthi R, et al. Global and regional burden of stroke during 1990–2010: findings from the Global Burden of Disease Study 2010. Lancet. 2014;383(9913):245–254.

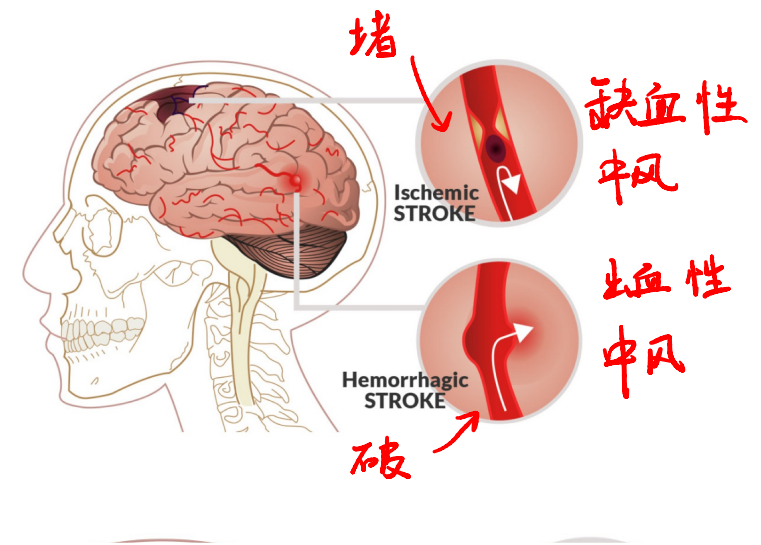
2. Suhyun, Lee, Yumi, et al. Effect of Virtual Reality-based Bilateral Upper Extremity Training on Upper Extremity Function after Stroke: A Randomized Controlled Clinical Trial[J]. Occupational Therapy International, 2016, 23(4):357–368.

3. Giovanni M, Stefano P, Andrea C, et al. Robot-assisted gait training for stroke patients: current state of the art and perspectives of robotics[J]. Neuropsychiatric Disease & Treatment, 2017, 13:1303–1311.

中风的类型

TYPES OF STROKE

- **Stroke** 中风定义：中枢神经系统（脑、脊髓、视网膜）内任何缺血或出血，无论是否有临床可检测症状。
 - Any ischemia or hemorrhage within the central nervous system, including brain, spinal cord, or retina regardless the presence of clinically detectable symptoms
- **Types of stroke**
 - Ischemic: caused by reduced blood supply
 - Hemorrhagic: caused by disrupted blood supply
 - Transient ischemic attack (TIA):
 - a transient episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia, without acute infarction



三种类型：

- 缺血性中风：供血减少（血管堵塞）——占大多数
- 出血性中风：供血被破坏（血管破裂出血）
- 短暂性脑缺血发作（TIA）：局灶性脑/脊髓/视网膜缺血导致的短暂神经功能障碍，无急性梗死（小中风，预警信号）

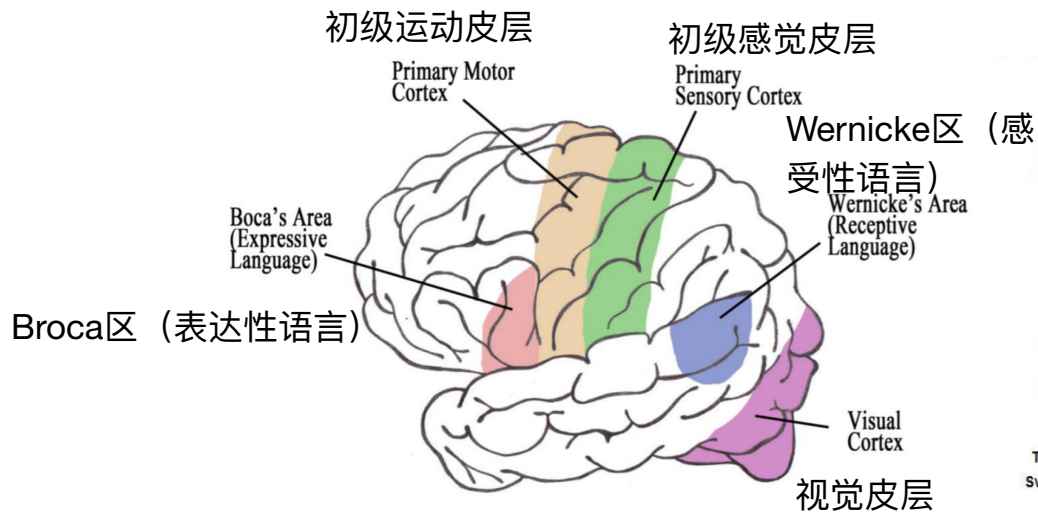


中风的后果

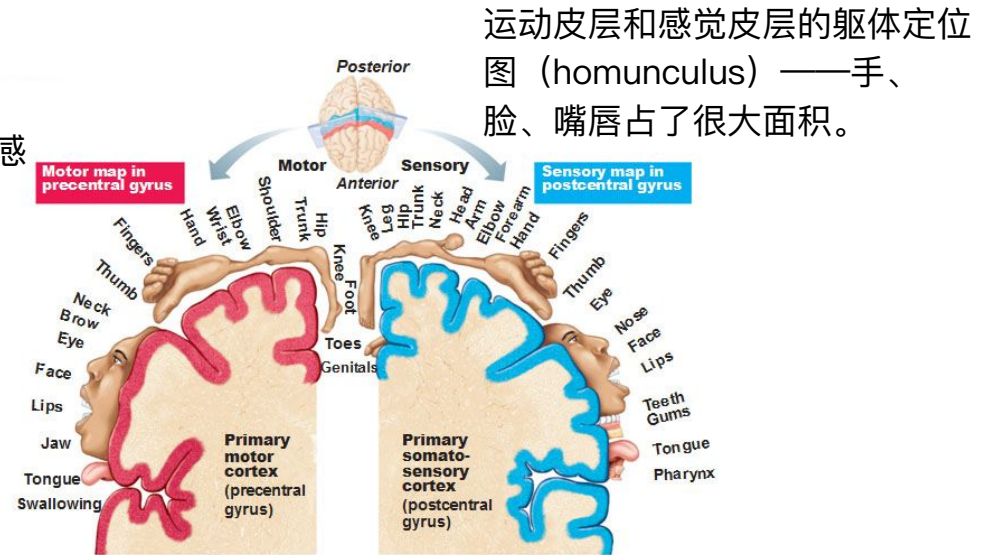
CONSEQUENCES OF STROKE

CONSEQUENCES OF STROKE

- Lesion location determines the consequences 损伤位置决定后果



图：大脑功能分区



运动皮层和感觉皮层的躯体定位图 (homunculus) ——手、脸、嘴唇占了很大面积。
中风损伤哪个脑区，就对应丢失哪部分功能。运动皮层管对侧身体运动，左侧脑中风导致右侧偏瘫。

四大功能障碍

分别对应运动皮层、Wernicke区、Broca区、视觉皮层/顶叶的损伤。

CONSEQUENCES OF STROKE

- **Motor function**
 - Inability to move one or more limbs on one side of the body
 - 运动功能：身体一侧一个或多个肢体无法运动
- **Cognitive function**
 - Inability to understand
 - 认知功能：无法理解
- **Speech function**
 - Inability to formulate speech
 - 言语功能：无法组织言语表达
- **Visual function**
 - Inability to see one side of the visual field (neglect)
 - 视觉功能：无法看到视野的一侧（忽略症）

运动功能损伤

MOTOR FUNCTION IMPAIRMENTS

- **Hemiparesis:**
 - 轻偏瘫 (Hemiparesis) : 肌肉无力或部分瘫痪, 局限于身体一侧 (上下肢)
 - muscular weakness or partial paralysis restricted to one side of the body (upper and lower limbs)
 - **Spasticity:**
 - a condition in which a muscle involuntarily contracts
 - Spastic co-contraction: inappropriate antagonist recruitment triggered by volitional command
 - **Abnormal motor synergies:**
 - loss of independent control of selected muscle groups (e.g., voluntary forward reaching with shoulder abduction and elbow flexion)
 - 异常运动协同 (Abnormal motor synergies) : 失去对特定肌群的独立控制 (例如想向前伸手, 肩膀外展和肘屈曲不由自主一起发生)
- 痉挛 (Spasticity) :
 - 肌肉不自主收缩
 - 痉挛性共同收缩: 自主指令触发了不该激活的拮抗肌



中风恢复机制

STROKE RECOVERY MECHANISM

STROKE RECOVERY

统计数据

Statistics:

- 10% spontaneous recovery in 8 – 12 weeks
- 10% severe debilitating, no significant recovery
- 80% with deficits, benefit from intense rehabilitation program

• 10%在8-12周内自发恢复

• 10%严重致残，无显著恢复

• 80%有功能缺损者，能从强化康复中获益

10%自己好，10%好不了，80%需要康复训练——这就是康复机器人的目标人群。

STROKE RECOVERY MECHANISMS

- **Spontaneous – no rehab** • 自发恢复（不需康复）：轴突和树突侧支出芽，建立新连接
 - Collateral sprouting of axons and dendrites to make new connections

- **Neural Plasticity** • 神经可塑性：神经通路和突触的重新连接
 - Reconnection of neural pathways and synapses

Once stroke occurs rehabilitation offers best opportunity for improving functional outcomes

• 一旦中风发生，康复提供改善功能预后的最佳机会

- **Why Stroke rehab works?**
 - Rehab facilitates Neuroplasticity
 - Animal and clinical evidence proved the efficacy

- 为什么中风康复有效？
 - 康复促进神经可塑性
 - 动物和临床证据证明了其疗效

神经可塑性

NEUROPLASTICITY

- 韦氏医学词典定义：活脑和神经系统的神经通路和突触，在经历损伤后持续改变的能力

- Definition in (Merriam-Webster medical dictionary)

- The capacity for continuous alteration of the neural pathways and synapses of the living brain and nervous system in response to experience of injury

- Neuroplasticity after CVA , TBI, and SCI investigated with

- Brain imaging techniques
- Animal models

- 中风（CVA）、创伤性脑损伤（TBI）、脊髓损伤（SCI）后的神经可塑性，通过脑成像技术和动物模型研究

- Neuroplasticity is much more extensive than previously thought

- 神经可塑性比以前认为的广泛得多

- Neuroplasticity actually occurs every day of our lives, as our nervous system changes in response to experience

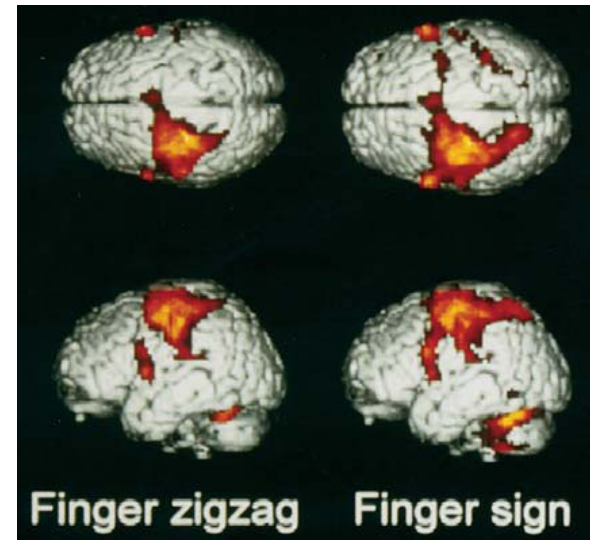
- 神经可塑性其实每天都在发生——我们的神经系统在经验中不断变化

以前认为大脑成年后就定型了，现在知道大脑终身都在"重新布线"——这就是康复的科学基础。

神经可塑性研究

NEUROPLASTICITY RESEARCH

- Brain motor areas which are “not used” can generate upper limb movements if properly stimulated
 - 大脑中“未使用”的运动区域，如果适当刺激，可以产生上肢运动
- The same motor function can be activated by multiple (different, non contiguous) brain motor areas
 - 同一个运动功能可以由多个（不同的、不相邻的）脑运动区激活



This situation proves the flexibility of the motor output organization and is a key issue to promote functional recovery after stroke • 这证明了运动输出组织的灵活性，是促进中风后功能恢复的关键

Motor learning and plasticity can be exploited for neurorehabilitation therapy, i. e. to promote functional recovery

- 运动学习和可塑性可以被利用来做神经康复治疗，即促进功能恢复

利用原则:

- 训练可以"激活"未受损脑区, 接管受损部分的功能
 - 大脑根据传入和传出信息的变化, 重新组织或"重新布线"自身
- 利用神经可塑性的原则:
 - 治疗时机
 - 任务特异性与一致性
 - 强度与重复
 - 患者动机与参与
 - 反馈

NEUROPLASTICITY

- Therapy training can "turn on" undamaged areas of the brain to take over function of damaged portion
 - Brain reorganize or "rewire" itself in response to changes in afferent and efferent information
- Principles in utilizing neuroplasticity:
 - Timing of therapy
 - Task specific and consistency
 - Intensity and repetition
 - Patient motivation and participation
 - Feedback

习得性废用

LEARNED NONUSE

- 未使用的身体系统会发生细胞、代谢和结构变化，时间久了中枢神经系统就把这种状态当成了"正常"

- 习得性废用导致肢体恢复和功能下降

- Unused body systems undergo cellular, metabolic, and structural changes which over time are interpreted by the CNS as the norm
- Learned nonuse results in decreased recovery and function of the limb

- **CONSTRAINT INDUCED MOVEMENT THERAPY (CIMT)**

强制性使用运动疗法 (CIMT)

- Technique used to reverse learned nonuse
 - The unimpaired upper extremity is restrained, forcing the patient to use the affected extremity
- 用来逆转习得性废用的技术
 - 把健侧上肢约束起来，强迫患者使用患侧上肢

中风后患者发现患手动不了，就学会了只用好手——时间长了，患手真的"被大脑放弃"了。CIMT就是把好手绑起来，逼大脑重新学习用患手。

用进废退

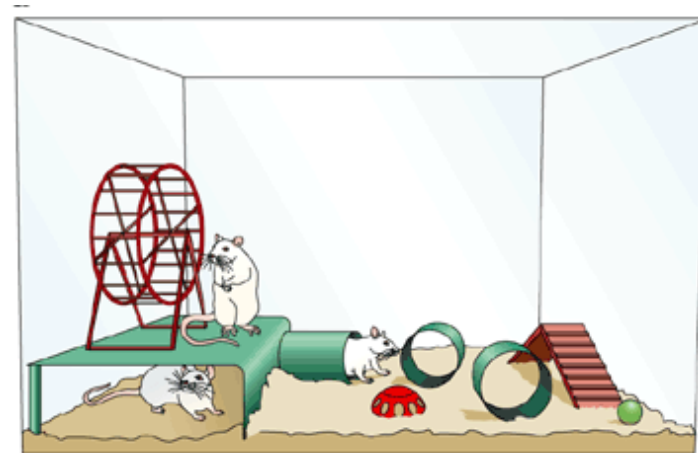
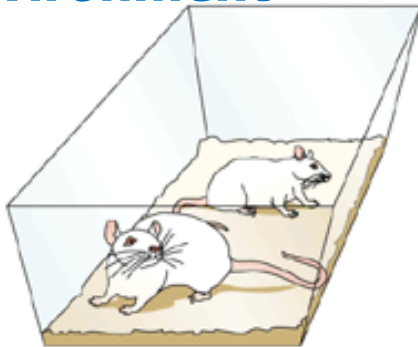
USE IT OR LOSE IT

• 康复训练（丰富环境+动物社交）增加皮层表征，带来后续功能恢复

Rehab training (enriched environments with animals) increases cortical representation with subsequent functional recovery

In animal studies key factors promoting recovery include *increased activity and complex, stimulating environment*

• 动物研究中，促进恢复的关键因素：活动量增加 + 复杂刺激环境



• 缺乏康复会减少皮层表征，延迟恢复

Lack of rehab decreases cortical representation and delays recovery

治疗时机

TIMING OF THERAPY

- 中风早期大脑"准备好"恢复
- 存在一个时间窗口——大脑对康复治疗反应最大，延迟不利于恢复
 - 尽早让患者挑战更高阶任务
 - 中风患者可能可以"跳级"完成传统实践中规定的低阶活动

- Brain is “primed” to “recover” early in post-stroke period
- There is a time window when brain is “primed” for maximal response to rehab therapies, such that delays are detrimental to recovery
 - Challenging patients to perform higher-order tasks as early as possible
 - Stroke patients may be able to leap-frog over lower-level activities prescribed by current traditional practice.

The Earlier the Better 越早越好.

治疗强度的作用

ROLE OF INTENSITY OF THERAPY

- Post-stroke rehab increases motor brain reorganization, while lack of rehab reduces reorganization
- More intensive motor training further increases brain reorganization
- Clinically, greater intensity of stroke rehab therapies is associated with improved outcomes

- 中风后康复增加运动脑区重组，缺乏康复则减少重组

- 更强的运动训练进一步增加脑重组

- 临床上，更强的中风康复强度与更好的预后相关

More is Better

越多越好。

中风康复必须是任务特异性的

STROKE REHAB MUST BE TASK-SPECIFIC

- Functional reorganization of cortex greater for tasks meaningful to animal; repetitive activity not enough
- Rehab must be task-specific, focusing on tasks important and meaningful to patient

• 对动物有意义的任务，皮层功能重组更大；单纯重复活动不够

• 康复必须是任务特异性的，聚焦于对患者重要和有意义的任务



不能让患者空挥手臂，要练他们真正需要的动作——拿杯子、开门、走路。

不活动与孤独

INACTIVE AND ALONE

In a conventional therapeutic day 传统治疗日中的时间分配：

- >50% time in bed
 - 28% sitting out of bed
 - 13% in therapeutic activities
 - Alone for 60% of the time
- >50%时间躺在床上
 - 28%坐在床外
 - 13%在做治疗活动
 - 60%时间独自一人

(Inactive and alone, Bernhardt et al, Stroke 2004)

Increased activity and environmental stimulation is important to neurological recovery

增加活动量和环境刺激对神经恢复很重要

现实很骨感——中风住院患者一半以上时间躺床上，真正训练只有13%，还没人陪。这就是为什么需要机器人来增加训练量。

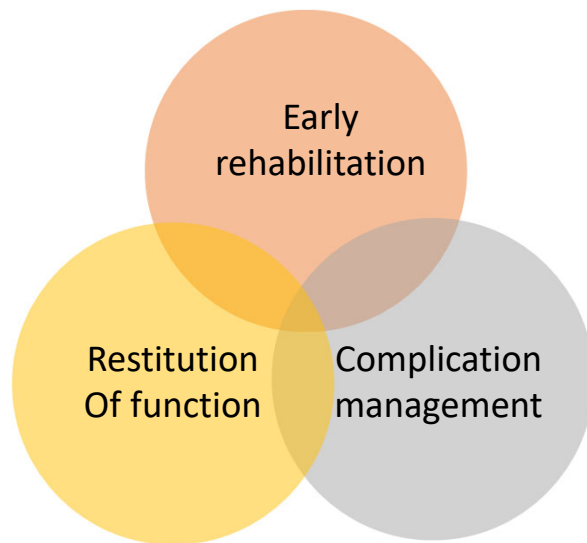


中风康复的临床实践与评估.

CLINICAL PRACTICES OF STROKE REHABILITATION AND ASSESSMENT

中风后康复的目标、

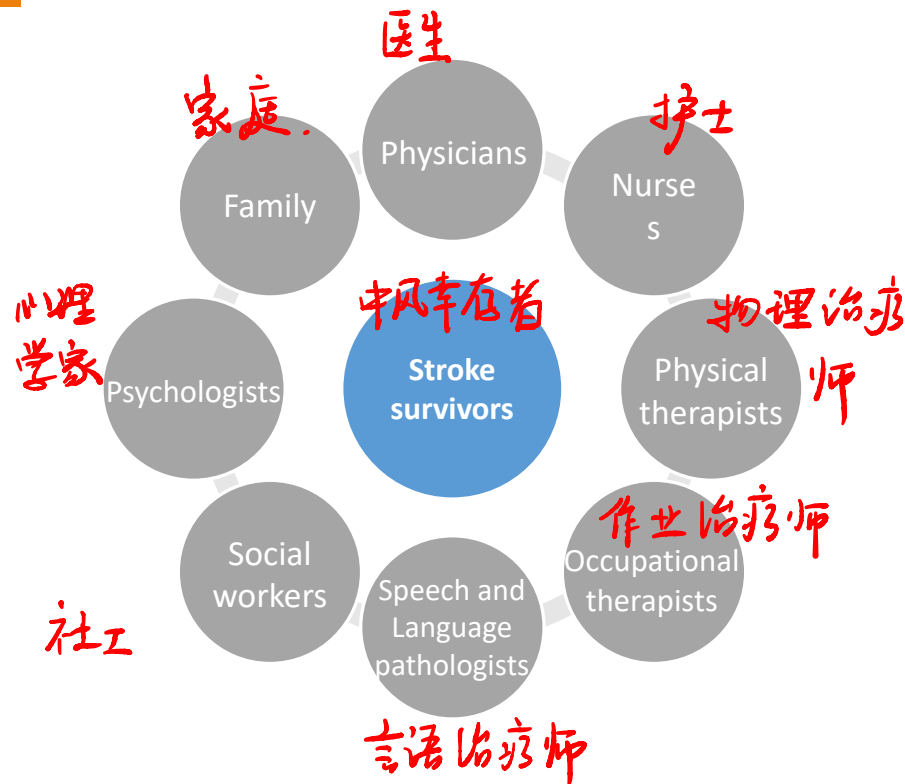
GOALS FOR POST-STROKE REHABILITATION



- Early rehabilitation
 - mobilization, mobility
 - Complication management
 - vital sign control, secondary stroke/complication prevention
 - Restitution of function
 - achieve the greatest possible autonomy and complete psychosocial reintegration
-
- 早期康复：动员、活动能力
 - 并发症管理：生命体征控制、预防二次中风/并发症
 - 功能恢复：实现最大程度的自主和完全的社会心理回归

中风后康复：团队协作

POST-STROKE REHABILITATION: TEAMWORK



- Physical factors
 - Cognitive effects
 - Physical effects
- Emotional factors
 - Motivation
 - Mood
- Social factors
 - Support
- Therapeutic factors
 - Early rehabilitation

- 身体因素：认知影响、身体影响
- 情绪因素：动机、情绪
- 社会因素：支持
- 治疗因素：早期康复

OT练日常生活活动（吃饭穿衣做饭），PT练运动能力（走路平衡力量）。

作业治疗 v.s. 物理治疗.

OCCUPATIONAL VS. PHYSICAL THERAPY

Occupational Therapy

- A client-centered health profession concerned with promoting health and well being through occupation



Physical Therapy

- Provides services to develop, maintain and restore maximum movement and functional ability throughout the lifespan



作业治疗（OT）：以患者为中心的健康专业，通过"作业活动"促进健康和福祉

物理治疗（PT）：提供服务以发展、维持和恢复整个生命周期的最大运动和功能能力

评估.

EVALUATION

- Motor deficits
 - Spasticity
 - Muscle strength
 - Motor control
 - Coordination
 - Motor quality
 - Sensory deficits
 - Perceptual deficits e.g. neglect
 - Cognitive deficits
 - Daily function
- 运动缺损:
 - 痉挛
 - 肌力
 - 运动控制
 - 协调
 - 运动质量
 - 感觉缺损
 - 知觉缺损 (如忽略症)
 - 认知缺损
 - 日常功能
- Language problem
 - Psychological problem
 - Social interaction problem
 - 语言问题
 - 心理问题
 - 社交互动问题

循证实践

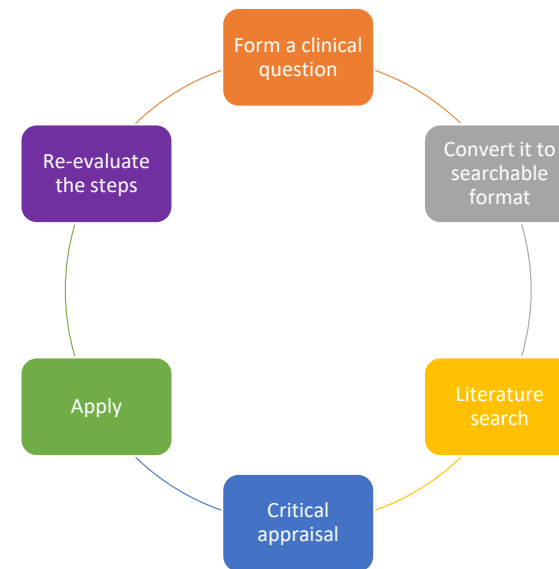
EVIDENCE-BASED PRACTICE

- The application of the principles of evidence-based medicine to clinical questions in the field of rehabilitation



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- 将循证医学原则应用于康复领域临床问题
- 三要素：临床专业知识 + 最佳研究证据 + 患者价值观与偏好
- 循环步骤：形成临床问题 → 转为可检索格式 → 文献检索 → 严格评价 → 应用 → 重新评价步骤



中风后运动康复 — 运动学习

POST-STROKE MOTOR REHABILITATION

- Motor learning: a set of processes associated with practice or experience leading to relatively permanent changes in the capabilities of responding
 - Acquisition phase: immediate effects
 - Retention phase: short-/long-term effects
 - Transfer test: ability to generalize to new task
- 运动学习：与练习或经验相关的一系列过程，导致反应能力的相对持久改变
 - 获得阶段：即时效果
 - 保持阶段：短期/长期效果
 - 迁移测试：泛化到新任务的能力

中风康复结果

STROKE REHABILITATION OUTCOMES

- 80% Independent Mobility
 - 70% Independent Personal Care
 - 40% Outside Home
 - 30% Work
- 80%恢复独立行走能力
 - 70%恢复独立个人护理
 - 40%能出门活动
 - 30%能重返工作

越往高级功能走，恢复比例越低——走路容易，上班难。

影响康复结果的因素

FACTORS AFFECTING REHAB OUTCOMES

- Neurological Deficits
 - Motivation Level
 - Learning Ability
 - Level of Emotional and Social Support
 - Coping and Adaptability
 - Medical Comorbidities
 - Rehabilitation Training
- 神经缺损程度
 - 动机水平
 - 学习能力
 - 情绪与社会支持水平
 - 应对与适应能力
 - 合并症
 - 康复训练

运动功能评估量表.

ASSESSMENTS OF MOTOR FUNCTION

- For both upper and lower extremities

- Fugl-Meyer Assessment (FMA)
- Motor Assessment Scale (MAS)
- Rivermead Motor Assessment (RMA)
- Chedoke-McMaster Stroke Assessment (CMSA)
- Motor Club Assessment (MCA)
- Motricity Index (MI)
- Stroke Rehabilitation Assessment of Movement (STREAM)

上下肢通用：

- Fugl-Meyer评估 (FMA) —— 最常用
- 运动评估量表 (MAS)
- Rivermead运动评估 (RMA)
- Chedoke-McMaster中风评估 (CMSA)
- 运动俱乐部评估 (MCA)
- 运动指数 (MI)
- 中风康复运动评估 (STREAM)



中风康复干预手段

STROKE REHABILITATION INTERVENTIONS

运动功能康复流程

MOTOR FUNCTION REHABILITATION

急性期/亚急性期:

关节活动度练习
体位摆放
功能性移动训练
ADL训练
夹板固定
痉挛管理

Acute/Subacute stage

- Range of motion exercise
- Positioning
- Functional mobility training
- ADL training
- Splinting
- Spasticity management



Chronic stage

- Continuum of acute/subacute phases
- Neurodevelopmental techniques
- Motor learning approaches
- Functional, task-specific overground locomotor training
- Task-oriented training

慢性期:

急亚急性期延续
神经发育技术
运动学习方法
功能性任务特异性训练
地面行走训练
任务导向训练



Community

- Home adaptations
- Assistive devices
- Supportive services
- Home exercise program

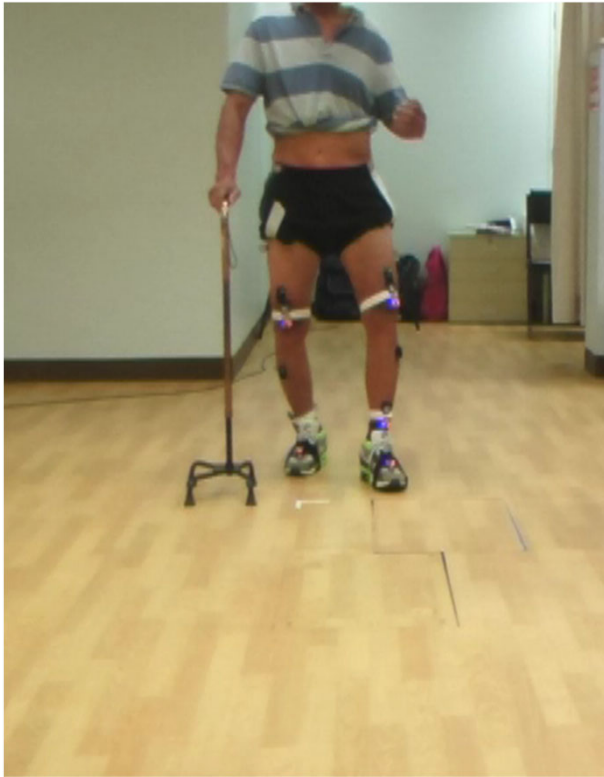


社区:

居家改造
辅助装置
支持服务
家庭运动计划

中风后异常步态

IMPAIRED GAIT AFTER STROKE



- Trunk compensation 躯干代偿
- Slow and asymmetric steps 步速慢且不对称
- Circumduction 画圈步态
- Reduced weight bearing on the affected leg 患腿负重减少
- Spastic gluteal and quadriceps muscle group 臀肌和股四头肌痉挛
- Weak hip flexors, hamstring, and ankle dorsiflexors → extended hip and knee & drop-foot 髋屈肌、腘绳肌、踝背屈肌无力 → 髋膝伸直+足下垂

步态康复目标

GOALS OF GAIT REHABILITATION



- Restoration of the ability to walk
 - Trunk control
 - Balance
 - Motor control
 - Muscle performance (strength, power, and endurance)
 - Independent ambulation
- 恢复行走能力
 - 躯干控制
 - 平衡
 - 运动控制
 - 肌肉表现（力量、爆发力、耐力）
 - 独立行走

传统步态康复

CONVENTIONAL GAIT REHABILITATION

关节活动度练习



Range of motion exercise

神经发育技术



Neurodevelopmental techniques

ADL训练



ADL training



Balance training
平衡训练



Overground gait training
地面行走训练



Spasticity management/splinting
痉挛管理/夹板固定

传统方法的局限

LIMITATIONS FOR CONVENTIONAL APPROACHES



- Additional manpower
- Inconsistent assistance
- 需要额外人力
- 帮助不一致



- Non-quantitative assessment
- 非定量评估

- Unable to track progression
 - Less effective
 - Inefficient feedback
 - Insufficient support
 - Inconsistent assistance
 - Labor intensive
- 无法跟踪进展
 - 效果较差
 - 反馈低效
 - 支持不足
 - 帮助不一致
 - 人力密集

这页是机器人介入的理由——传统康复靠治疗师人力，训练量不够、不一致、没法定量评估。

行走康复治疗.

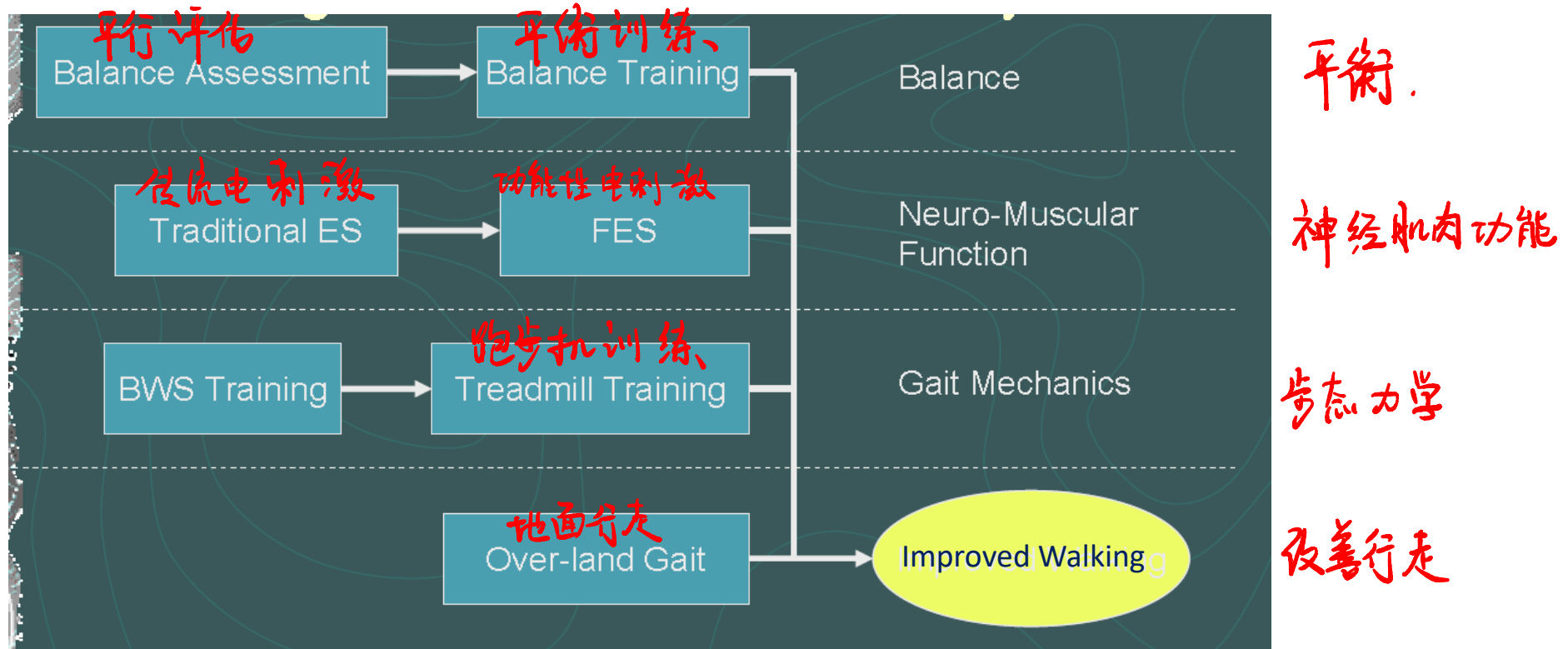
LOCOMOTION REHABILITATION THERAPY

当前实践的关键组成:

- **Key components of current practice:**
 - Balance assessment and training
 - Functional Electrical Stimulation (FES)
 - Body Weight Support (BWS) gait training
 - BWS treadmill gait training
- 平衡评估与训练
- 功能性电刺激 (FES)
- 减重步行支持 (BWS) 步态训练
- 减重跑步机步态训练

行走康复治疗

LOCOMOTION REHABILITATION THERAPY



BALANCE ASSESSMENT AND TREATMENT

平衡训练可以：

- Balance training allows:
 - Comprehensive evaluation of transitional and precursor movements required for normal walking.
 - Retraining of identified deficiencies.
 - Enable evidence-based training
 - 综合评估正常行走所需的过渡性和前驱性动作
 - 对已识别的缺陷进行再训练
 - 实现循证训练



Balance Master

平衡测试
系统

功能性电刺激

FUNCTIONAL ELECTRICAL STIMULATION (FES)

- Electrical stimulation 电刺激分类：
 - Sensory electrical stimulation • 感觉电刺激
 - High-frequency transcutaneous electrical nerve stimulation (TENS): electrical somatosensory stimulation of a peripheral nerve via cutaneous electrodes (80 – 100 Hz) ◦ 高频经皮神经电刺激 (TENS)：通过皮肤电极对周围神经进行电躯体感觉刺激 (80–100 Hz)
 - Electroacupuncture ◦ 电针
 - Muscle electrical stimulation • 肌肉电刺激
 - Low-frequency transcutaneous electrical nerve stimulation: 1 – 5 Hz
 - Neuromuscular electrical stimulation (NEMS): induces muscle (10 – 50 Hz)
- Functional Electrical Stimulation (FES)
 - A technique that causes a muscle to contract through the use of an electrical current.
 - 功能性电刺激 (FES)：通过电流使肌肉收缩的技术

FES的证据

EVIDENCE OF FES

Upper Limbs

- Improve hand function and dexterity, motor control, motor function, functional independence, & strength

上肢：改善手功能和灵巧性、运动控制、运动功能、功能独立性和肌力



http://www.neurolinkphysio.co.uk/images/img_arm_myotrac1.jpg

下肢：对运动控制、运动功能、痉挛、步行效率、步速、肌力有效

Lower Limbs

- Effective on motor control, motor function, spasticity, walking efficiency, walking speed, strength

下肢功能性电刺激 (FES)

LOWER EXTREMITY FUNCTIONAL ELECTRICAL STIMULATION (FES)

- **Bioness Leg Unit:**
 - Miniature Gait sensor
 - Wireless Control Unit
 - Facilitate a more fluid gait
 - Increased walking speed

Bioness腿单元 (L300) :

- 微型步态传感器
- 无线控制单元
- 促进更流畅的步态
- 提高步速



Bioness L300 Plus系统概览

BIONESS L300 PLUS SYSTEM OVERVIEW



减重支撑 训练

BODY WEIGHT SUPPORT (BWS) TRAINING

- BWS benefits:
 - Reducing patient effort
 - Increased training intensity
 - Improve postural retraining
 - Research validated the training

BWS的好处: efficacy

- 减少患者用力
- 增加训练强度
- 改善姿势再训练
- 研究验证了训练疗效



LiteGait BWS system

中风后上肢功能障碍

IMPAIRED UPPER LIMB FUNCTION AFTER STROKE



左图：患者尝试用患手抓罐子，动作笨拙。

- Muscle weakness
- Sensory loss
- Pain
- Spasticity
- Abnormal motor synergies
- 肌无力
- 感觉丧失
- 疼痛
- 痉挛
- 异常运动协同

上肢康复目标

GOALS OF UPPER LIMB REHABILITATION



- Prevent complications
 - Immobilization
 - Muscle flaccidity
 - Shoulder subluxation
 - Improve motor/sensory functions
 - Return to family/community
- 预防并发症
 - 制动
 - 肌肉软瘫
 - 肩关节半脱位
 - 改善运动/感觉功能
 - 回归家庭/社区

上肢功能恢复关键组成

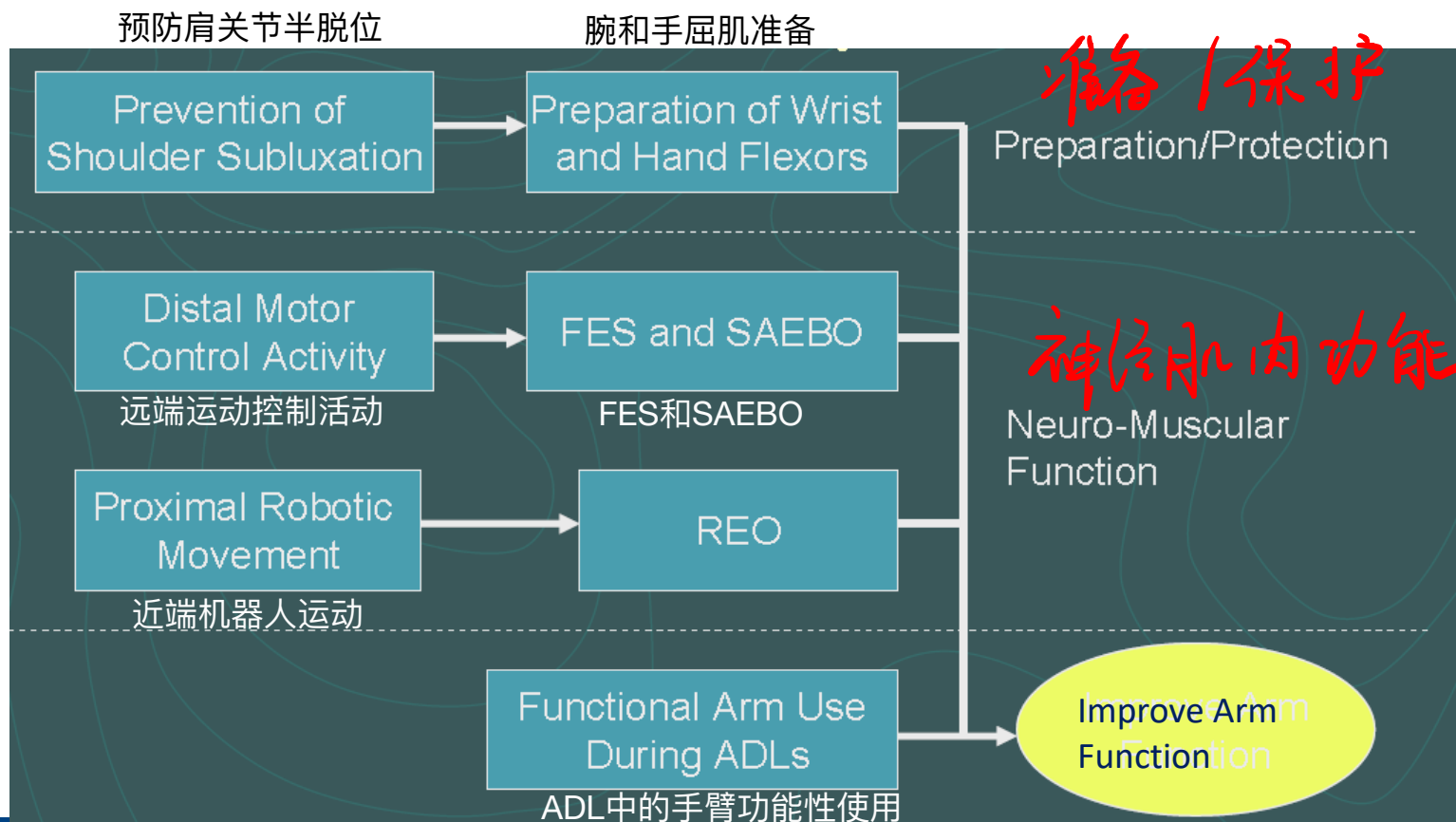
UPPER LIMB FUNCTION RECOVERY

- Key Components

- Joint and soft tissue protection and preparation
- Repetitive stimulation of proximal motor control
- Active facilitation of distal motor control
- FES assessment and treatment
 - 关节与软组织保护和准备
 - 近端运动控制的重复刺激
 - 远端运动控制的主动促进
 - FES评估与治疗

手臂功能恢复

ARM FUNCTION RECOVERY



改善手臂功能.

准备与保护

ARM FUNCTION RECOVERY -PREPARATION AND PROTECTION

- To provide:
 - Reduction of subluxation in sitting and standing
 - Reduction of shoulder pain
 - Compression of shoulder, elbow and wrist during standing and ambulation to facilitate return

- 坐位和站位时减少半脱位
- 减轻肩痛
- 站立和行走时压迫肩、肘、腕，促进回归
- 研究证明疗效

• Research proved efficacy



GivMohr Slings

吊带

ARM FUNCTION RECOVERY -PREPARATION AND PROTECTION

- Addresses soft tissue shortening of the long finger flexors
 - The most limiting factor to regaining hand function following a stroke.
- Utilizes a low load long duration stretch to return the fingers to the desired position.

SAEBOstretch (伸展支具) :

- 解决手指长屈肌的软组织短缩
 - 这是中风后手功能恢复的最大限制因素
- 利用低载荷长时间拉伸，将手指恢复到期望位置



The SAEBOstretch

动态手部矫形器

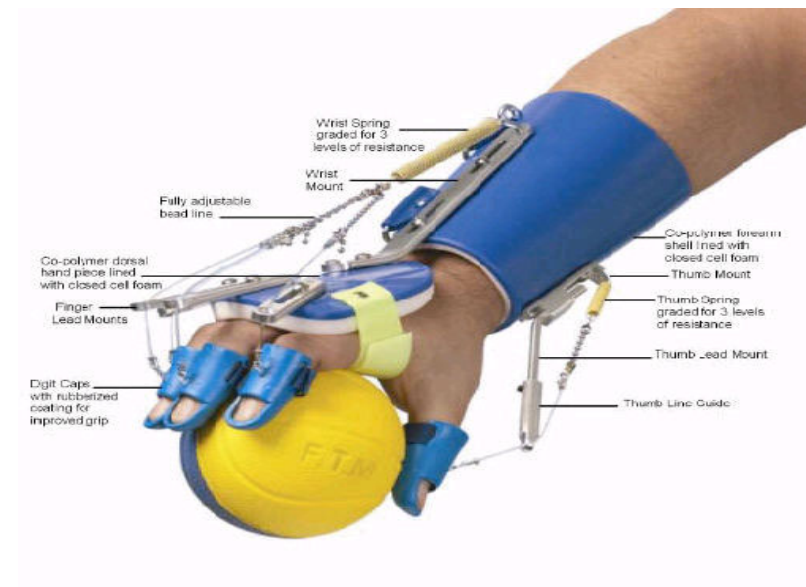
ARM FUNCTION RECOVERY

- Dynamic hand orthosis example:

- Spring loaded extension system
- Extension spring system assists in re-opening the hand to release the object
- Assists in re-opening thumb and fingers following functional grasping
- Non-electrically based, purely mechanical

- 弹簧加载伸展系统
- 伸展弹簧系统辅助重新张开手以释放物体
- 在功能性抓握后辅助重新张开拇指和手指
- 非电动，纯机械

SaeboFlex



www.saebo.com

上肢FES

UPPER EXTREMITY FES

- **Bioness H200 Arm Unit:**
 - The can be used both as a neuroprosthesis (i.e., as a functional aid for performing activities of daily living) and as a therapeutic device to aid motor recovery of function following a stroke.

Bioness H200手臂单元：

- 既可作为神经假肢（即作为完成日常生活活动的功能辅助）
- 也可作为治疗装置，辅助中风后功能的运动恢复





中风康复的新兴解决方案

EMERGING SOLUTIONS OF STROKE
REHABILITATION

虚拟现实

VIRTUAL REALITY (VR)

- An artificial environment which is experienced through sensory stimuli (as sights and sounds) provided by a computer and in which one's actions partially determine what happens in the environment.
- 人工环境，通过感官刺激（视觉、声音）体验，由计算机提供，人的动作部分决定环境中发生什么



https://hscweb3.hsc.usf.edu/wp-content/uploads/2014/07/HCM7559_RSS.jpg

右图：患者在CAVE沉浸式VR中，站在跑步机上走虚拟林间小路。

VR 的证据

EVIDENCE OF VIRTUAL REALITY

上肢:

- 对手臂功能有显著效果
- 对改善日常功能有益
- → 最有效的VR方案
- → 患者特征

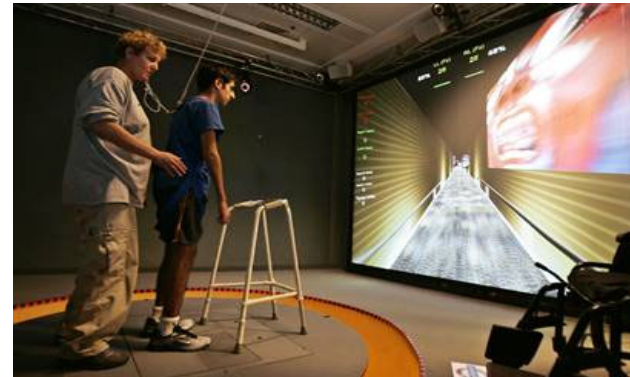
Upper Limbs

- Significant effect on arm function
- Beneficial in improving daily function
- Most effective VR program
- Patient characteristics



Lower Limbs

- No significant effect on gait speed
- VR + conventional therapy is more effective in improving balance and gait ability
- Participation level
- Follow-up



下肢:

- 对步速无显著效果
- VR+常规疗法在改善平衡和步态能力方面更有效
- → 参与度
- → 随访

Virtual Reality for hand rehab

节律性治疗 — 交互式节拍器

RHYTHMIC THERAPY- INTERACTIVE METRONOME

- Computerized program with headphones
 - Computer generated reference tone
 - Match the rhythmic beat with repetitive motor actions (hands or feet)
 - Immediate feedback measured in milliseconds
 - Score can be recorded as feedback
- 带耳机的计算机程序
 - 计算机生成参考音调
 - 用重复运动动作（手或脚）匹配节律节拍
 - 以毫秒为单位即时反馈
 - 分数可记录作为反馈

www.interactivemetronome.com

交互式节拍器 (机制)

INTERACTIVE METRONOME

Mechanisms:

- Addresses brain timing, rhythmicity, and synchronicity
 - Increases the speed and coordination of informational signals within the brain
 - Improves the processing abilities that affect attention, motor planning, and sequencing
 - Strengthens many fundamental learning capacities and higher level cognitive skills such as planning, organizing, and language
- 针对脑内计时、节律性和同步性
 - 提高脑内信息信号的速度和协调性
 - 改善影响注意力、运动规划和序列编排的处理能力
 - 增强许多基础学习能力和高阶认知技能，如规划、组织和语言

步态训练、音乐辅助治疗

机器人辅助治疗 (RT)

ROBOT-ASSISTED THERAPY (RT)

- Using robots to facilitate recovery or compensate impairments
 - Therapeutic robots
 - End-effector-type
 - Exoskeleton-type
- Upper limbs vs. lower limbs
 - 用机器人促进恢复或补偿缺损
 - 治疗机器人：
 - 末端执行器型 (end-effector)
 - 外骨骼型 (exoskeleton)
 - 上肢 vs 下肢

末端执行器上肢



http://www.imalab.net/sites/default/files/image/news/MITRobot_406X248.jpg

上肢外骨骼



<http://greaterkansas.mdnews.com/sites/default/files/media/vc-armeospringjanarombeck750.jpg>



<http://www.reha-stim.de/cms/assets/images/GT-I-mit-Person>

下肢外骨骼跑步机



http://www.inovis.ch/en/wp-content/uploads/sites/2/2016/03/stories_Hocoma_Lokomat_bgwhite.jpg

上肢外骨骼

机器人技术对中风的好处

BENEFITS OF ROBOTICS TECHNOLOGIES FOR STROKE

1. Provide high-dosage, high-intensity practice
2. Provide highly consistent practice that helps reduce variability
3. Reduce burden on occupational and physical therapists
4. Provide a mechanism for quantitative evaluation

1. 提供高剂量、高强度练习
2. 提供高度一致的练习，帮助减少变异性
3. 减轻作业治疗师和物理治疗师的负担
4. 提供定量评估的机制

这四条是康复机器人的核心价值——对应传统康复“量不够、不一致、靠人力、没法定量”的四大痛点。

Summary

- Definition & Epidemiology of stroke
- Consequences of stroke
- Stroke recovery Mechanism
- Clinical practices of stroke rehabilitation and assessment
- Stroke Rehabilitation Interventions
- Robotics in stroke rehabilitation

总结

- 中风的定义与流行病学
- 中风的后果
- 中风恢复机制
- 中风康复的临床实践与评估
- 中风康复干预手段
- 中风康复中的机器人技术